MIL-T-23824A(OS)
9 August 1971
Superseding
MIL-T-23824(Wep)
15 July 1963

MILITARY SPECIFICATION

TORPEDO BEACON LIGHT

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the minimum requirements for the procurement of a Torpedo Beacon Light, NAVORD Drawing 695202, which is used in the exercise section of the Torpedo Mk 37 Mods 0, 2, and 3 to aid in the postrun location and recovery of the torpedo at the end of a night-time exercise run. This torpedo beacon light is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Reservation Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1355

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DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 620156	Shipping (Packing) Condition
NAVORD Dwg 695202	Torpedo Beacon Light Outline Drawing

(Copies of specifications, standards, and drawings required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publication

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The torpedo beacon light consists of two main assemblies: a power pack and a flashhead.

3.1.1 Power pack. The power pack is cylindrically shaped, measuring 10-3/8 inches long and 4 inches in diameter, and is designed for interior mounting at the lower center of the exercise section of Exercise Head Mk 64 Mod 0. Functionally, the power pack increases the 30 vdc input, from the torpedo battery, to a sufficient voltage to operate the flashtube of the flashhead assembly.

3.1.2 Flashhead. Components of the flashhead are assembled within a housing designed for installation in the handhole cover opening of the torpedo exercise section. When activated at the end of an exercise run, the beacon light flashes on and off at l-second intervals. The flashing light is visible, from a small recovery craft, at a range of approximately 6000 yards.

3.2 General requirements

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3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the beacon light shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining prior to starting production, that the contractor's production methods are capable of producing beacon lights that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Material. All material and components used in the fabrication of the torpedo beacon light shall be of a quality that will insure that the torpedo beacon light will meet the performance and environmental requirements specified herein.

3.2.4 Marking. Marking of the torpedo beacon light shall be in accordance with NAVORD Dwg 695262.

3.2.5 Design

3.2.5.1 Accessibility. The design of the torpedo beacon light shall provide for the convenient and efficient replacement of the lens and flashtube of the flashhead assembly and all power pack components.

3.2.5.2 Polarity. The design of the torpedo beacon light shall provide for the isolation of all circuits from their packaging to 'permit operation with systems employing either a positive or negative ground.

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28.

3.2.7 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Definitions. Definitions applicable to the environmental requirements of this specification are given in MIL-T-82590.

3.4 Performance requirements and product characteristics. The torpedo beacon light shall meet the following performance requirements and product characteristics:

3.4.1 Size. The torpedo beacon light shall meet the dimensional requirements of NAVORD Dwg 695202.

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3.4.2 Weight. The torpedo beacon light shall weigh 7.5 pounds maximum.

3.4.3 Polarity reversal. The torpedo beacon light shall not be damaged by polarity reversal at the input terminals.

3.4.4 Power requirements. The torpedo beacon light shall meet the functional requirements of this specification when energized by the following source:

Input voltage - 30 volts dc plus or minus 5 volts
Input power - 6 watts plus or minus 10 percent

3.4.5 Functional

3.4.5.1 Flash rate. The torpedo beacon light shall flash at a rate of 60 plus or minus 15 flashes per minute.

3.4.5.2 Flashtube output. The output of the flashtube shall be not less than 30,000 beam candle power.

3.4.6 pressure. The lens and surrounding surfaces of the flashhead assembly (see Dwg 695202) shall be capable of withstanding an external hydrostatic pressure of 550 psig plus or minus 10 psig for a period of 30 minutes, without leakage or deformation.

3.4.7 Operating life. The torpedo beacon light shall be capable of continual satisfactory operation for a period of 250 hours.

3.5 Environmental requirements. The beacon light shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5..

3.5.1 Vibration. The beacon light shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The beacon light shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.5.1.2 Vibration, operating. The beacon light shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.5.2 Temperature and humidity

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3.5.2.1 High temperature and humidity, nonoperating. The beacon light shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.1.

3.5.2.2 High temperature, operating. The beacon light shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.2.

3.5.2.3 Low temperature, nonoperating. The beacon light shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.3.

3.5.2.4 Low temperature, operating. The beacon light shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.4.

3.6 Cleanup. Prior to and after final assembly all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as excessive rosin flash shall be removed.

3.7 Workmanship. The torpedo beacon light, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The workmanship exhibited in the preproduction sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 classification of inspections. Inspection of the beacon light shall be classified as follows:

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(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five beacon lights, manufactured in accordance. with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the beacon light by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate '*inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five beacon lights will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The beacon lights shall be considered satisfactory if they meet all specifications and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each beacon light delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the torpedo beacon light shall be subjected to quality conformance tests under the following conditions.

4.4.1.1 Temperature. Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18.33 degrees centigrade (C) to 35.0 degrees C).

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4.4.1.2 Altitude. Normal ground.

4.4.1.3 Vibration. None.

4.4.1.4 Humidity. Room ambient to 95 percent relative (maximum).

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified it shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 Light meter: Edgerton, Germeshausen, and Grier, Inc., a two-unit system of; Model #560 Lite-Mike and Model #561 Detector Head, or equivalent.

4.4.3.2 Power source as specified in 3.4.4 and equipped with a SPST switch.

4.4.3.3 Stop watch or similar timing device.

4.4.3.4 Hydrostatic test fixture capable of applying an external hydrostatic pressure of 550 psig to the lens and surrounding surface of the flashhead assembly including the O-ring. The external O-ring on the beacon light shall affect the seal within the fixture. The fixture diameter in the area of the O-ring seal shall be 4.003 - 4.005.

4.4.4 Test procedure. The torpedo beacon light shall be tested and inspected in accordance with the following procedures:

4.4.4.1 Inspection. The contractor shall perform all necessary inspection procedures to ensure that the torpedo beacon light meets the requirements of 3.2.3 through 3.2.6, 3.4.1, 3.4.2, and Section 5 of this specification.

4.4.4.2 Functional test

4.4.4.2.1 Pressure. Secure the torpedo beacon light flashhead assembly in the test fixture (4.4.3.4) and apply a hydrostatic pressure of 550 plus or minus 10 psig to the lens and surrounding surfaces (see NAVORD Dwg 695202) for a period of 5 minutes. There shall be no evidence of leakage or deformation.

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4.4.4.2.2 Flash rate and flash tube output

4.4.4.2.2.1 Remove flashhead assembly from test fixture and connect to power pack.

4.4.4.2.2.2 Connect power pack to power source (4.4.3.2) through SPST switch.

4.4.4.2.2.3 Simultaneously throw switch on and start stopwatch (4.4.3.3). Over a 5-minute period, determine flash rate and measure flashtube output, using light meter (4.4.3.1). The torpedo beacon light shall meet the requirements of 3.4.5.

4.4.4.2.3 polarity reversal. Reverse the input leads to the power pack and throw SPST switch on and then off. Return input leads to original positions, and repeat the procedures specified in 4.4.4.2.2. The torpedo beacon light shall meet the requirements of 3.4.5.

4.4.4.3 Visual and dimensional examinations. Each beacon light shall be examined to determine conformance with the requirements for workmanship and dimensions,

4.4.4.4 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Reproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The beacon light, less pack or packaging, shall be mounted in a manner simulating actual installation in the torpedo and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The beacon light shall be subjected to the “Vibration, nonoperating” test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

4.5.1.3 Vibration, operating. The beacon light shall be subjected to the “Vibration, operating” test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.5.1.2.

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4.5.2 Temperature and humidity

4.5.2.1 High temperature and humidity, nonoperating. The beacon light shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.1.

4.5.2.2 High temperature, operating. The beacon light shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.2.2.

4.5.2.3 Low temperature, nonoperating. The beacon light shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.3.

4.5.2.4 Low temperature, operating. The beacon light shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.2.4.

4.5.3 Operating life. After completion of all other preproduction tests, the torpedo beacon light shall be operated and periodically tested for satisfactory operation until a total of 250 hours operating time has elapsed (including operating the elapsed during the performance of the tests under 4.5.1 through 4.5.2.4) and then shall again pass the quality conformance tests of 4.4 and operate satisfactorily. Test conditions during the operating life test may be any condition, or combination of conditions of ambient temperature, humidity, and marine environment that are specified in the other tests. Operating time shall be considered as time during which the torpedo beacon light is actually performing its intended use, and does not include time during which it is in "ready" or "standby" condition.

5. PREPARATION FOR DILIVERY

5.1 Reservation and packaging

5.1.1 Level A

5.1.1.1 Cleaning. Unless otherwise specified, the torpedo beacon light assembly exterior surfaces only shall be cleaned in accordance with Process C-1 or MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the torpedo beacon light assembly shall be dried in accordance with one or more of the procedures of MIL-P-116, provided the procedure is not injurious to item.

5.1..1.3 Preservative application. None.

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5.1.1.4 Unit packaging. Unit packaging of the torpedo beacon light assembly shall be in accordance with NAVORD LD 620156.

5.1.1.5 Intermediate package. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning and drying. Cleaning and drying shall be in accordance with 5.1.1.

5.1.3.2 Preservative application. None.

5.1.3.3 Unit packaging. Unit packaging shall be in accordance with Method III of MIL-P-116. Unit package shall contain one torpedo beacon light assembly.

5.1.3.4 Intermediate package. Not applicable.

5.2 Packing

5.2.1 Level A

5.2.1.1 One torpedo beacon light assembly shall be packaged as described in 5.1.1.4.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The torpedo beacon light assembly, packaged in accordance with 5.1.3.3, shall be packed to afford protection against damage during direct shipment from supply source to first receiving activity for immediate use. Shipping containers shall conform to the rules and regulations applicable to mode of transportation.

5.3 Marking

5.3.1 Special marking. The following special marking shall appear on the top, on one side, and on one end of each container:

DELICATE INSTRUMENT
HANDLE WITH CARE

5.3.2 Normal marking. In addition to marking required by contract or project order, unit packages and shipping containers shall be marked in accordance with requirements of MIL-STD-129, including serial numbers.

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6. NOTES

6.1 Intended use. The torpedo beacon light is intended for use in Torpedo Mark 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging.

(d) Required level of packing.

(e) Assigned activity for preproduction and periodic production inspections l

(f) Data required other than that specified in 3.2.l.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

Preparing Activity
Navy - OS
Project No. 1355-N273

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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